

NCR MANAGER · QUALITY MANAGEMENT SYSTEM

Performance Test Plan

Scalability targets, design principles and the load-test scenarios that validate latency at enterprise scale.

DOCUMENT

PERFORMANCE TEST PLAN

VERSION

1.0

ISSUED

June 27, 2026

PLATFORM

Kinetic

CONFIDENTIAL · CONTROLLED DOCUMENT

| Targets

- **Users:** 10,000 active, 100,000 supported (named).
- **Data:** millions of submissions; large evidence collections; 12 sites, global.
- **UX latency:** list page < 1s P95; record-360 drawer < 1.5s P95; dashboard tiles < 3s P95 (cached).

| Design principles already in place

- **Indexed, selective KQL** — every filter field is indexed; compound indexes for common pairs.
- **25-record client pages** — no client ever loads more than one page; keyset/cursor paging only.
- **Bounded server aggregation** — dashboard collects are page-bounded and de-duplicated, then cached 5 minutes with in-flight de-duplication.
- **No N+1 in lists** — list rows render from one query; record-360 fans out a fixed, bounded set of indexed queries (not per-row calls).
- **Append-only history not loaded in lists** — audit trail is queried only inside a record drawer, filtered by `Record Type + Record ID` (compound index).

Test scenarios

#	SCENARIO	LOAD	PASS CRITERIA
P-1	Module list filtered query	500 rps, 5M-row form	P95 < 800ms; 0 errors; constant memory
P-2	Record-360 fan-out	100 rps	P95 < 1.5s; query count fixed regardless of links
P-3	Dashboard cold vs warm	50 concurrent	cold < 8s, warm (cached) < 500ms; single upstream burst (in-flight dedup)
P-4	Web API list pagination	200 rps, deep pages	stable latency across pages; no growth from token depth
P-5	Idempotent intake	100 rps with replays	replays return cached result; no duplicate submissions
P-6	Concurrent edits + audit trail	50 rps writes	each write yields exactly one trail entry per changed field; no lost updates
P-7	Escalation/timer sweep (workflow)	1M open records	scheduled job completes within window; no UI impact
P-8	Soak	24h at 30% peak	no memory growth, no connection leak, stable latency

Instrumentation

- Correlation IDs end-to-end; capture per-endpoint P50/P95/P99 and upstream Kinetic query counts.
- Track cache hit ratio on `/dashboard` and `/v1/metrics`.
- Assert query plans use indexes (no full submission scans) via platform query metrics.

Scaling levers

- Raise/lower dashboard collect `maxPages` per form size; push >N-page analytics to the data-platform integration rather than the live dashboard.
- Add compound indexes for any new high-cardinality filter pair before enabling it in the UI.
- Horizontally scale the app server (stateless except the in-memory cache; safe to run N replicas — cache is per-replica and short-TTL).

Exit criteria

All scenarios meet pass criteria at target load for two consecutive runs; soak shows no degradation; no unindexed scans observed. Results recorded as PQ evidence in `VALIDATION-PACKAGE.md` §4.